

TOP 10 ARTICLES IN CERTIFICATION RESEARCH

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Former Chief of Assessments
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- ABMS Scholars Program
- Research Grant Program
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 - Research Preconference – ABMS Conference 2024-2026
 - Procedural Skill Assessment – Consensus Conferences 2025, 2026



OVERALL METHODS

- 10 ...because it's a round number
- Disclaimers:
 - Unscientific procurement method
 - Disproportionate (beneficial) influence of Lars Peterson at the ABFM and Tom Granatir ABMS
- No Disclosures
- More Opinion.....Less Detail

Article #1:

Vandergrift, Ladon, Weng, Gray.

JAMA Internal Medicine | [Original Investigation](#) | [LESS IS MORE](#)

Low-Value Services and Longitudinal Knowledge Assessment Performance

Jonathan L. Vandergrift, MS; Bruce E. Landon, MD; Weifeng Weng, PhD; Bradley M. Gray, PhD

[+ Editor's Note](#)

[+ Supplemental content](#)

IMPORTANCE Low-value services (LVSs) remain common and account for a substantial amount of wasteful US health care spending.

OBJECTIVE To assess whether LVS use is associated with physicians' general clinical knowledge as measured by performance on the American Board of Internal Medicine's Longitudinal Knowledge Assessment (LKA).

DESIGN, SETTING, AND PARTICIPANTS This cross-sectional study included general internists enrolled in the LKA in 2022 or 2023 caring for Medicare fee-for-service beneficiaries aged 67 years or older in an outpatient setting during their first year of participation.

EXPOSURE Physician first-year LKA scores.

MAIN OUTCOMES AND MEASURES Receiving any of 25 LVSs designed to measure low-value primary care. Patient level regressions were adjusted for patient demographics, comorbidities, physician practice characteristics, and patient location.

RESULTS Of 898 365 included patients, 59% were female with a mean (SE) age of 76.8 (0.02) years, and 30% received at least 1 LVS. Of 7089 physicians, 3568 (50%) were male with a mean (SE) age of 52.5 (0.1) years. Having a physician with a top vs bottom quartile LKA score was associated with a logistic-regression adjusted 7.9% lower (95% CI, -10.5% to -5.3%, $P < .001$) chance of receiving any LVS during the year (bottom: 31.0% [95% CI, 30.4% to 31.7%]; top: 28.6% [95% CI, 28.1% to 29.1%]); absolute difference: -2.5 percentage points (95% CI, -3.3 to -1.6, $P < .001$). Significant differences were observed by LVS subtype including diagnostic or preventive testing (bottom: 10.4% [95% CI, 9.8% to 11.0%], top: 8.7% [95% CI, 8.3% to 9.1%]; absolute difference: -1.7 percentage points [95% CI, -2.5 to -0.9]; $P < .001$), cancer screening (bottom: 13.8% [95% CI, 13.4% to 14.3%], top: 12.3% [95% CI, 11.9% to 12.7%]; absolute difference: -1.5 [-2.1 to -0.9], $P < .001$), and imaging (bottom: 13.8% [95% CI, 13.4% to 14.2%]; top: 13.2% [95% CI, 12.9% to 13.5%]; absolute difference: -0.6 [95% CI, -1.1 to -0.1]; $P = .02$). LKA score quartile differences for individual LVSs were most prominent for total and free triiodothyronine testing in patients with hypothyroidism and prostate-specific antigen testing in males 75 aged years and older.

CONCLUSIONS AND RELEVANCE The findings of this cross-sectional study suggest patients cared for by physicians with higher first-year LKA scores were less likely to receive LVSs.

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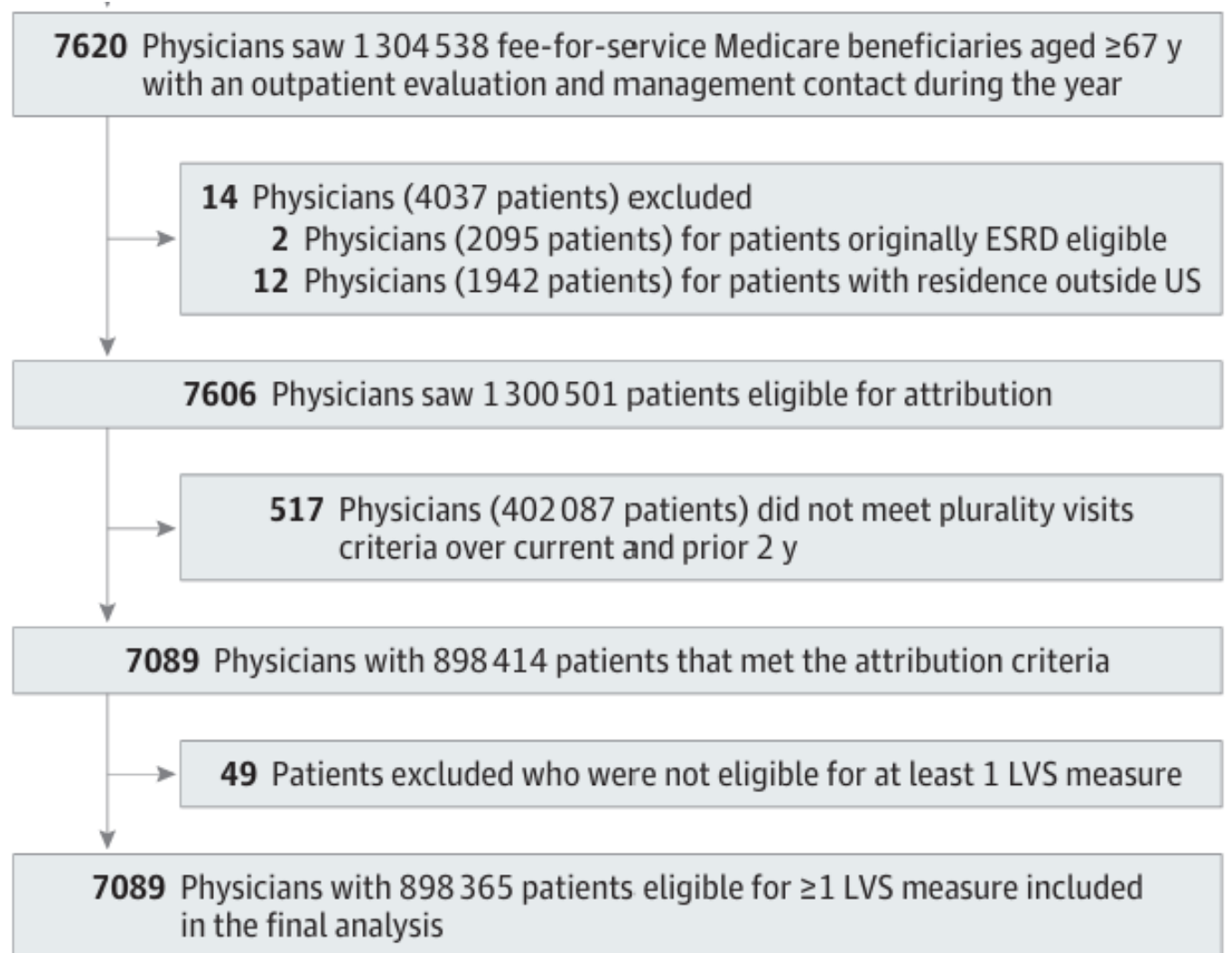
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CONCLUSIONS AND RELEVANCE The findings of this cross-sectional study suggest patients cared for by physicians with higher first-year LKA scores were less likely to receive LVSs.

The Why

- General clinical knowledge measured by LKA score
- Association with patient level Medicare Data
- Outcome
 - Were there errors of commission?
 - MD did something of low value

Methods



Results

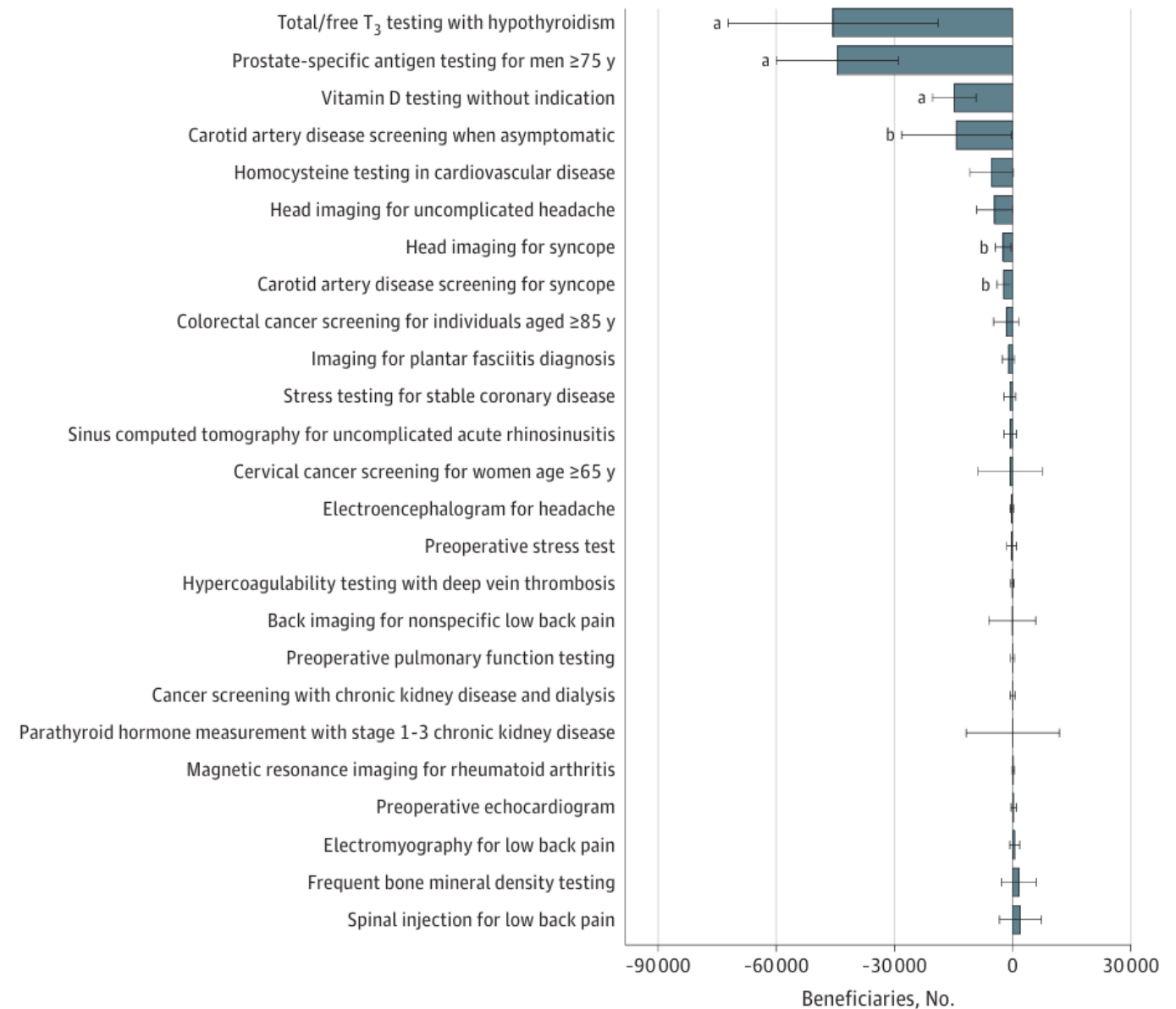
Table 2. Regression-Adjusted LVS Rates by LKA Score Quartile^a

LVS	Adjusted mean (95% CI)	Absolute difference (95% CI)	P value	% Difference (95% CI)	P value
Any LVS (n = 898 365; physician n = 7089)					
1 (Bottom)	31.0 (30.4 to 31.7)	1 [Reference]		1 [Reference]	
2	30.0 (29.4 to 30.6)	-1.0 (-1.9 to -0.1)	.03	-3.2 (-6.0 to -0.4)	.02
3	29.4 (28.9 to 29.9) ^b	-1.6 (-2.4 to -0.8) ^b	<.001 ^b	-5.1 (-7.7 to -2.5) ^b	<.001 ^b
4 (Top)	28.6 (28.1 to 29.1) ^b	-2.5 (-3.3 to -1.6) ^b	<.001 ^b	-7.9 (-10.5 to -5.3) ^b	<.001 ^b

Chance of receiving any Low Value Service during the year (e.g. 28.6% vs. 31.0%)

Implications

Figure 2. Bar-and-Whisker Plot of Patients Receiving Low-Value Service If All Clinically Active General Internists Matched the LVS Rate Observed for Top Quartile-Scoring Physicians



Key Takeaways

In this national study of physician knowledge and LVS use, we found that ambulatory patients cared for by general internists who scored better during their first year of LKA were significantly less likely to receive any LVSs. This supports the view that unexplained differences in LVS use between physicians in the same practice in part reflect differences in their clinical knowledge.^{12,13}

Gray et al (ABIM)

JAMA Internal Medicine August 2025 Volume 185, Number 8

Continuous Medical Knowledge Assessment Progress Report Score and Hospitalized Patient Outcomes

The American Board of Internal Medicine started offering a Longitudinal Knowledge Assessment (LKA) in 2022 as an alternative to the 10-year long-form Maintenance of Certification assessment. Participants answer 30 questions in an open-book format quarterly, and competency is assessed every 5 years. Physicians receive preliminary scores to identify knowledge gaps, but how these are associated with clinical outcomes is unclear. We measure associations between the first-year LKA score and hospitalized patient outcomes for the first 2 cohorts of participating hospitalists.

Methods | Our cross-sectional study, conducted among hospitalists enrolled in LKA in 2022 and 2023, examined associations between preliminary scores (hereafter, *scores*) and hospitalized patient outcomes 2 years prior to enrollment. The

 [Supplemental content](#)

Figure 2. Preliminary Score Quartile Associations: Adjusted Percentage Difference Compared With the Bottom Quartile Mean Outcome

Outcome	Adjusted mean (95% CI) ^a	Adjusted absolute difference (95% CI) ^a	P value	Difference, % (95% CI) ^b	Favors better outcome	Favors worse outcome	P value
Primary outcome measures							
7-d Mortality per 1000 hospitalizations (unadjusted mean, 50.3)							
Quartile 1	53.0 (50.9 to 55.1)	0 [Reference]	NA	0 [Reference]			NA
Quartile 2 vs quartile 1 (bottom)	49.1 (47.1 to 51.2)	-3.8 (-7.1 to -0.6)	.02	-7.3 (-13.2 to -1.3)			.02
Quartile 3 vs quartile 1 (bottom)	49.1 (47.1 to 51.2)	-3.9 (-7.1 to -0.6)	.02	-7.3 (-13.2 to -1.4)			.02
Quartile 4 (top) vs quartile 1 (bottom)	48.9 (46.4 to 51.4)	-4.1 (-7.7 to -0.5)	.03	-7.8 (-14.3 to -1.2)			.02



Impact of medical oncologist certification on survival outcomes in metastatic colorectal cancer: evidence from the SCRUM-Japan MONSTAR-SCREEN Observational Study

Kyosuke Seguchi¹ · Tadayoshi Hashimoto^{1,2} · Takao Fujisawa^{2,3} · Taro Shibuki^{2,4} · Chigusa Morizane⁵ · Norio Nonomura⁶ · Hiroji Iwata⁷ · Hidemichi Watari⁸ · Susumu Okano³ · Kenjiro Namikawa⁹ · Yoichi Naito¹⁰ · Hideaki Bando^{1,2} · Takayuki Yoshino^{1,2}

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Abstract

Background Board certification in surgical oncology improves cancer outcomes, but the impact of medical oncologist certification on systemic therapy outcomes is unknown. We investigated whether the Japanese Society of Medical Oncology (JSMO) board certification of the enrolling physician was associated with treatment quality and survival.

Methods This retrospective cohort study used data from the SCRUM-Japan MONSTAR-SCREEN observational study, a nationwide multicenter genomic screening program enrolling patients from 2019 to 2022. Standard treatment implementation rates were compared across cancer types as a secondary outcome. The primary outcome was overall survival (OS) in patients with metastatic colorectal cancer (mCRC) using Kaplan–Meier analysis and Cox proportional hazards regression with sequential adjustment for clinical and molecular covariates.

Results Among 2,184 patients, standard treatment rates were comparable between board-certified and non-board-certified groups (97.0% vs. 96.9%). In 358 patients with mCRC, enrollment by a board-certified physician was associated with longer OS (median, 36.5 vs. 30.7 months; hazard ratio (HR): 0.66, 95% CI 0.50–0.87; log-rank $p=0.003$) even after adjustment for age, sex, tumor sidedness, region, trial participation, molecular targeted therapy, metastatic organ count (HR, 0.73; 95% CI 0.55–0.97; $p=0.029$), and after additional adjustment for RAS and BRAF V600E status (HR, 0.64; 95% CI 0.47–0.88; $p=0.005$). Board-certified physicians used molecular targeted therapy more frequently (89.5% vs. 80.4%; $p=0.036$).

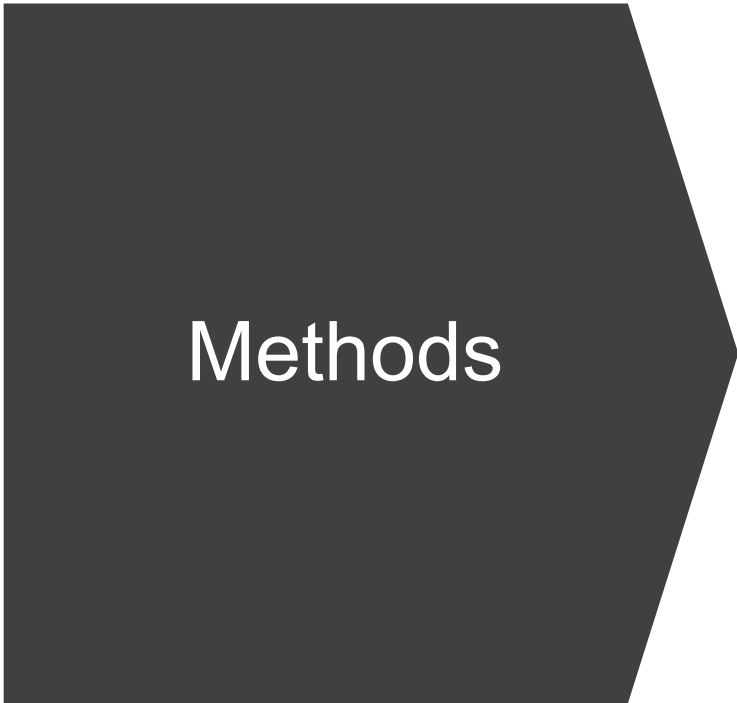
Conclusions JSMO board certification was independently associated with improved OS in mCRC despite comparable guideline adherence, suggesting that specialist expertise extends beyond treatment selection.

Keywords Medical oncologist · Board certification · Colorectal cancer · Chemotherapy · Targeted therapy · Guideline adherence

Article #2: Seguchi et al.

The Why

- Certification by the Japanese Society of Medical Oncology (JSMO)
- Disease specific registry – Metastatic Colon Cancer
- Outcome
 - Disease free survival
 - Overall survival

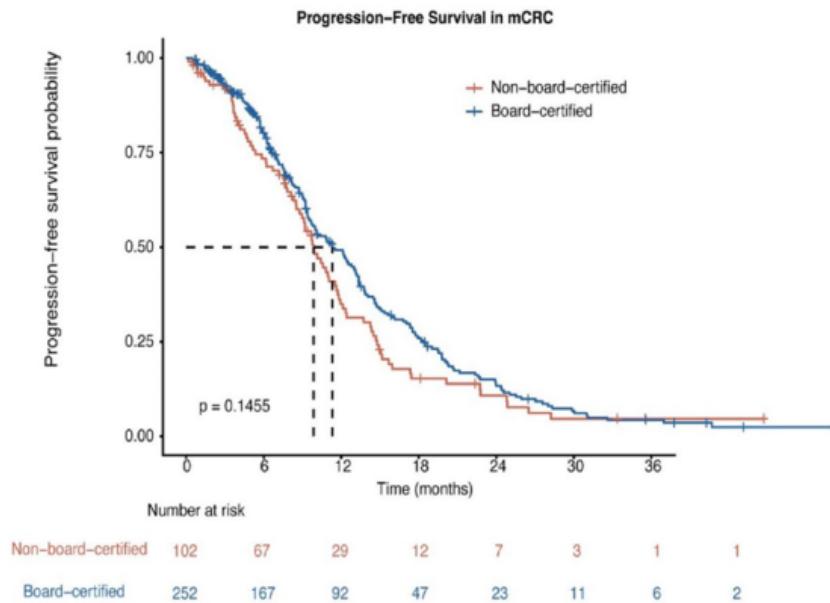


Methods

- Retrospective analysis of a prospectively collected cohort study
- Exposure: Board Certification
- Outcome:
 - Subset of patients with metastatic colorectal cancer
 - Treatment implementation
 - Survival rates

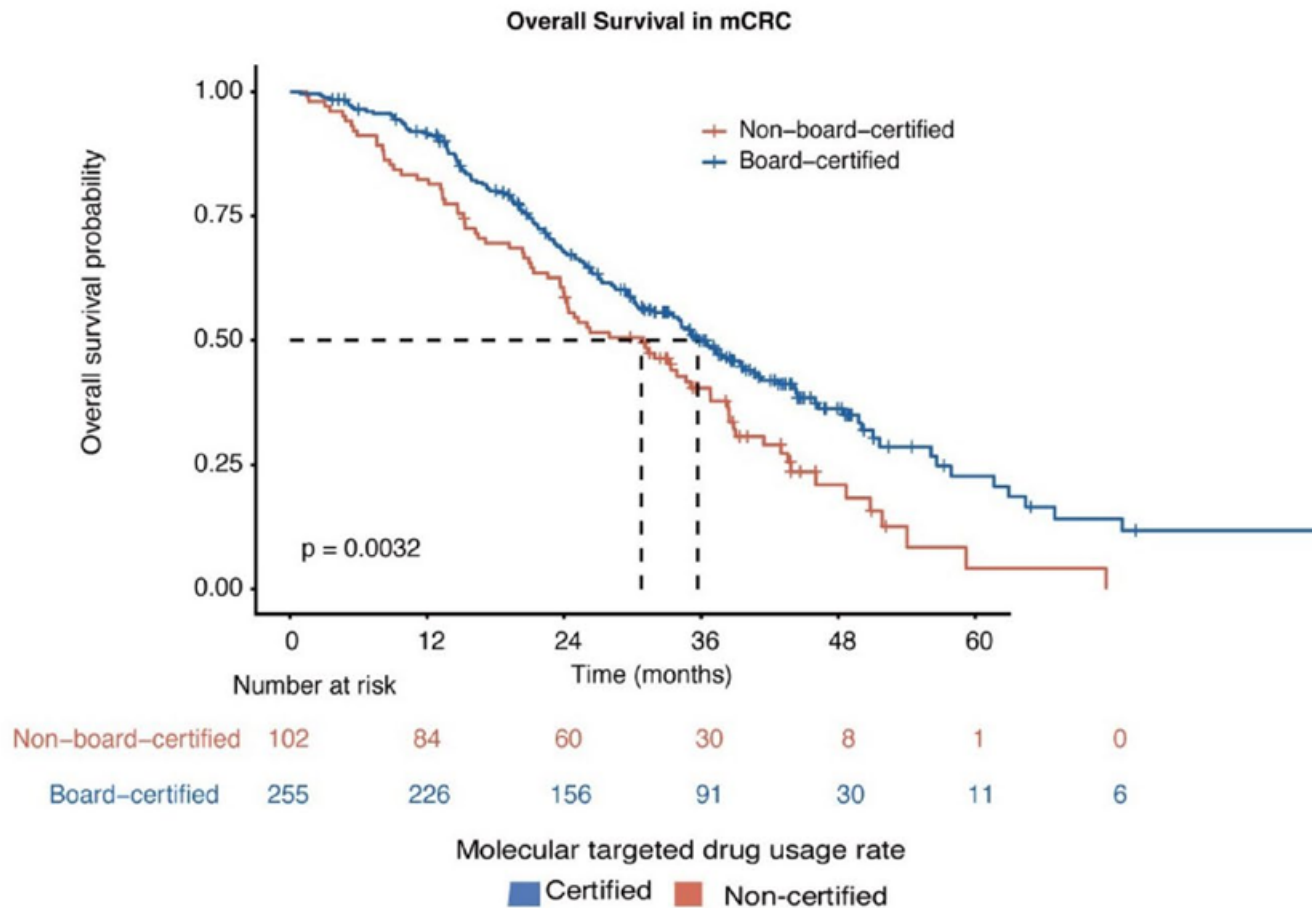
Results

A

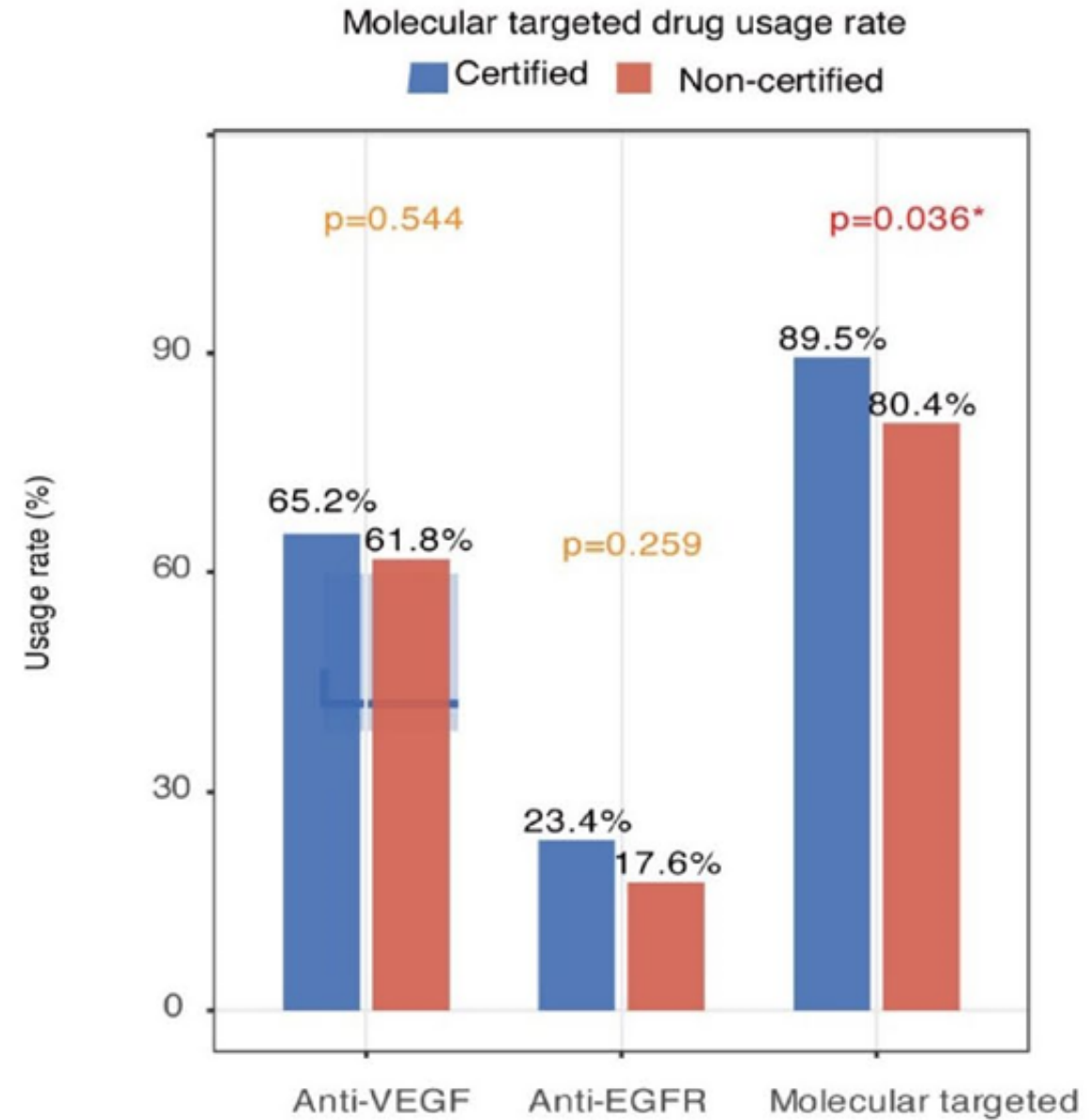


N=255 Certified; 102 Not Certified

3



Results



Key Takeaways

In this nationwide cohort study using the SCRUM-Japan MONSTAR-SCREEN observational study, enrollment by a JSMO board-certified medical oncologist was independently associated with a 27–34% reduction in the hazard of death among patients with mCRC, an association that persisted across sequential multivariable models. This survival advantage was observed despite virtually identical standard treatment implementation rates (97.0% vs. 96.9%), suggesting that the benefit of specialist involvement extends beyond guideline adherence.

Article #3: Gavarkovs et al.

Short Report

Physicians' Values as Motivators for Continuing Professional Development: A Cross-Sectional Study

Adam G. Gavarkovs, PhD; William McCauley, MD, MHPE

Introduction: According to Self-Determination Theory, when health professionals see their continuing professional development (CPD) efforts as aligned with their intrinsic values (eg, helping others), they may develop a more desirable motivational orientation toward CPD. The aim of this cross-sectional study was to investigate the effect of physicians' values on their CPD motivation.

Methods: Practicing physicians in a single Canadian province were administered an online survey that assessed their motivation for engaging in CPD, the importance they ascribe to three intrinsic and three extrinsic values, and the extent to which engaging in CPD activities is aligned with intrinsic and extrinsic values.

Results: A total of 122 physicians completed the survey. Seeing CPD activities as contributing to helping others was positively associated with intrinsic regulation, identified regulation, and introjected regulation. Seeing CPD activities as contributing to personal growth was positively associated with identified regulation.

Discussion: As predicted by the Self-Determination Theory, physicians who perceived CPD as aligned with intrinsic values tended to report a more desirable motivational orientation toward CPD. CPD providers can support health professionals' motivation by highlighting connections between CPD activities and intrinsic values in promotional materials.

Keywords: motivation, engagement, values, physicians, self-determination theory

DOI: 10.1097/CEH.0000000000000589

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The Why

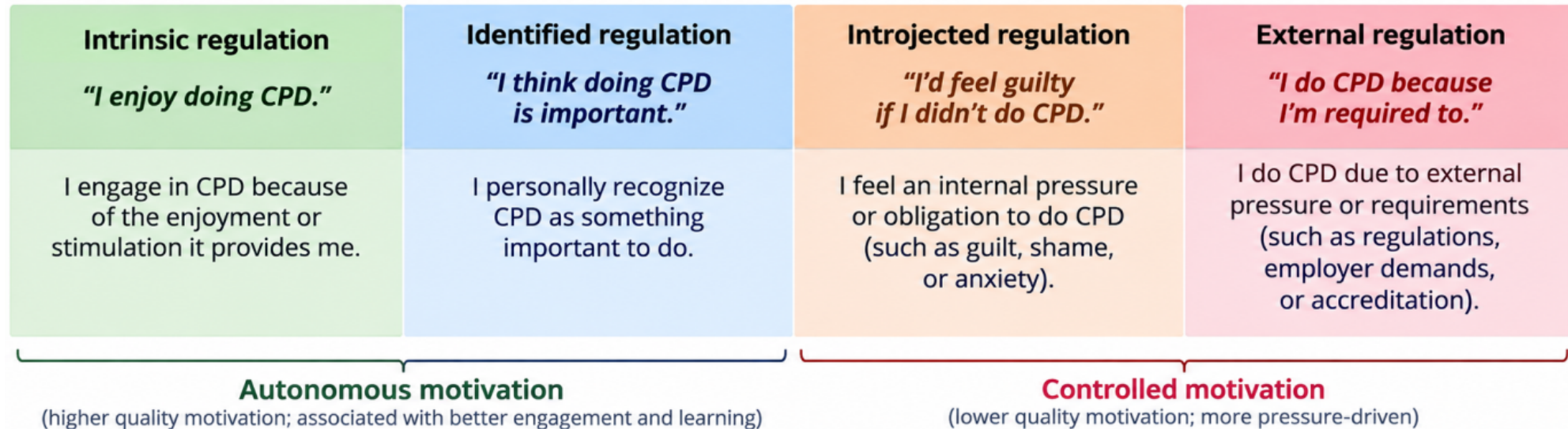
Self-Determination Theory assesses the degree to which a person acts based on Intrinsic versus Extrinsic Motivation.

A Continuum of Motivation for Continuing Professional Development (CPD)

Self-Determination Theory (SDT)

More autonomous
(SELF-DETERMINED)

More controlled
(EXTERNALLY DRIVEN)



Surveyed
122
Canadian
physicians

Supplementary Digital Appendix 2. Summary of participant characteristics.

Characteristic	Count (%)
Specialty	
<i>Family Medicine</i>	32 (27%)
<i>Emergency Medicine</i>	22 (19%)
<i>Surgery</i>	21 (18%)
<i>Internal Medicine</i>	18 (15%)
<i>Paediatrics</i>	10 (8%)
<i>Anaesthesiology</i>	2 (1%)
<i>Psychiatry</i>	2 (1%)
<i>Pathology & Laboratory Medicine</i>	2 (1%)
<i>Other</i>	9 (9%)
Practice context	
<i>Academic</i>	67 (55%)
<i>Community</i>	55 (45%)

Key Results

Predictors of Physicians' Motivation for Continuing Professional Development (CPD)

Bayesian ordinal regression results (standardized estimates with 95% HDI)

Predictor <i>(Value alignment and interaction terms)</i>	I do CPD because I enjoy it / find it stimulating. (Intrinsic regulation) <i>I engage in CPD because of the enjoyment or stimulation it provides me.</i>	I do CPD because I believe it is important. (Identified regulation) <i>I personally recognize CPD as something important to do.</i>	I do CPD because I would feel guilty, ashamed, or anxious if I didn't. (Introjected regulation) <i>I feel an internal pressure or obligation to do CPD.</i>	I do CPD because someone else or the situation requires me to. (External regulation) <i>I do CPD due to external pressure or requirements.</i>
	Estimate (95% HDI)	Estimate (95% HDI)	Estimate (95% HDI)	Estimate (95% HDI)
Alignment: Helping others (intrinsic value)	0.25 (0.04 – 0.46)	0.41 (0.17 – 0.64)	0.26 (0.04 – 0.50)	0.08 (-0.13 – 0.29)
Alignment: Close relationships (intrinsic value)	0.15 (-0.04 – 0.33)	-0.03 (-0.24 – 0.18)	0.15 (-0.04 – 0.34)	-0.10 (-0.29 – 0.08)
Alignment: Personal growth (intrinsic value)	0.13 (-0.09 – 0.34)	0.47 (0.23 – 0.72)	-0.13 (-0.35 – 0.09)	-0.04 (-0.25 – 0.17)
Alignment: Financial wealth (extrinsic value)	-0.05 (-0.23 – 0.14)	-0.03 (-0.23 – 0.17)	0.12 (-0.07 – 0.30)	0.10 (-0.08 – 0.28)
Alignment: Admired by people (extrinsic value)	0.02 (-0.17 – 0.21)	0.17 (-0.03 – 0.38)	0.18 (-0.01 – 0.37)	0.12 (-0.06 – 0.30)
Alignment: Appealing image (extrinsic value)	-0.00 (-0.20 – 0.20)	-0.23 (-0.45 – -0.00)	0.14 (-0.06 – 0.34)	0.06 (-0.13 – 0.25)

More autonomous
(SELF-DETERMINED)



More controlled
(EXTERNALLY DRIVEN)

Key Takeaways

The purpose of this study was to investigate the effects of health professionals' values on their motivation to engage in CPD. Consistent with SDT, physicians who perceived CPD as aligned with two intrinsic values—helping others and personal growth—tended to report more autonomous motivation to engage in CPD. Our results are consistent with SDT-informed studies that suggest framing a learning activity as serving intrinsic values (but not extrinsic values) can enhance autonomous motivation, deep learning strategy use, persistence, and performance.^{7,8}

The Ultimate Study?

More controlled
(EXTERNALLY DRIVEN)



More autonomous
(SELF-DETERMINED)

MOTIVATION Quartile

Figure 2. Preliminary Score Quartile Associations: Adjusted Percentage Difference Compared With the Bottom Quartile Mean Outcome

Outcome	Adjusted mean (95% CI) ^a	Adjusted absolute difference (95% CI) ^a	P value	Difference, % (95% CI) ^b	Favors better outcome	Favors worse outcome	P value
Primary outcome measures							
7-d Mortality per 1000 hospitalizations (unadjusted mean, 50.3)							
Quartile 1	53.0 (50.9 to 55.1)	0 [Reference]	NA	0 [Reference]			NA
Quartile 2 vs quartile 1 (bottom)	49.1 (47.1 to 51.2)	-3.8 (-7.1 to -0.6)	.02	-7.3 (-13.2 to -1.3)	—●—		.02
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Article #4: Wang, Bazemore, Newton, Price

Original Research

Spaced Repetition Enhances Self-Rated Learning Confidence: A Large Randomized Trial Among Practicing Family Physicians

Ting Wang, PhD; Zachary J. Morgan, MS; Andrew Bazemore, MD, MPH; Warren P. Newton, MD, MPH; David W. Price, MD

Introduction: Metacognitive accuracy—the extent to which individuals’ perceptions of their knowledge align with actual performance—is critical in clinical decision making. Overconfidence can lead to diagnostic errors, while insufficient confidence may delay decisions or prompt unnecessary testing, increasing health care costs. Although longitudinal knowledge assessments with spaced repetition have been shown to improve learning and knowledge transfer, their impact on metacognition remains underexplored. This study investigates changes in confidence under different spaced repetition strategies in a large cohort of family physicians.

Methods: Data were drawn from the American Board of Family Medicine’s Continuous Knowledge Self-Assessment, a longitudinal assessment administered quarterly. Participants rated their confidence on a 6-point scale after each question. In total, 16,751 physicians who completed assessments at baseline, quarter 6, and quarter 10 were included. Participants were randomized to control or intervention groups receiving repeated or clone questions at various intervals. One-sample *t* tests assessed confidence changes for questions initially answered incorrectly but subsequently answered correctly. Multivariate regression controls demographic factors.

Results: Confidence increased significantly for repeated questions, with moderate effect sizes in single repetition groups and large effect sizes in two-time repetition groups. Clone questions also showed gains, though smaller in magnitude. After adjusting for clinician demographics, male and younger physicians generally reported higher confidence.

Discussion: Spaced repetition enhances both learning and metacognitive accuracy. Two spaced repetitions yielded larger gains. These findings support incorporation of metacognitive-enhancing strategies into CME to improve clinician self-assessment and potentially reduce diagnostic error.

Keywords: metacognition, confidence, learning

DOI: 10.1097/CEH.0000000000000615

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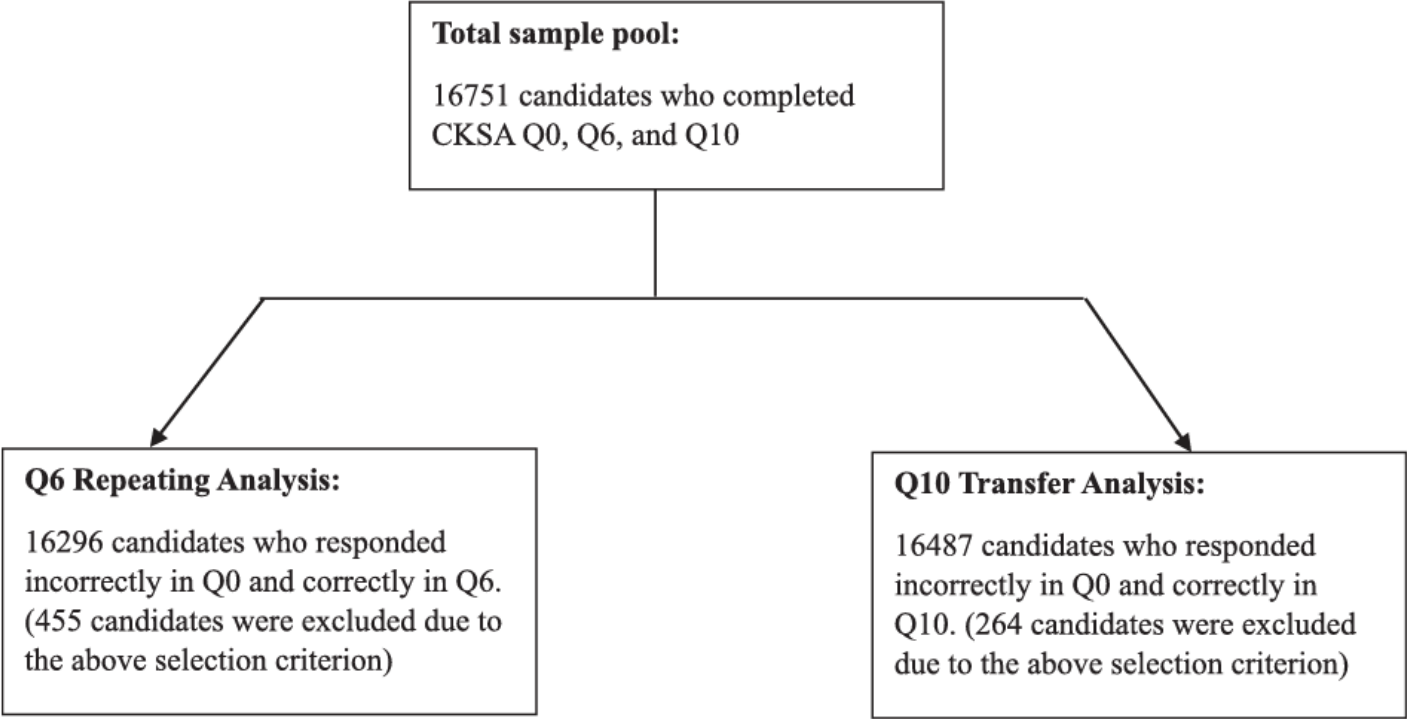
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The Why

- LKA context
- Spaced repetition works in terms of knowledge gains
- Does it work in terms of “calibration” = confidence in your answers?

METHODS



Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
X						✓				✓

← Spaced Repetition →

Confidence

Confidence

Results

Confidence: 6-point scale

TABLE 4.

Multivariable Model: Predictors of Confidence on Repeated Items (quarter 6 [Q6])

Variable	Estimate β (95% CI)	<i>P</i>
Intercept	3.16 (3.06, 3.27)	<.001
Group		
Control group	Ref.	
Group 1	0.05 (0.01, 0.09)	.012
Group 2	0.09 (0.05, 0.13)	<.001
Group 3	0.11 (0.07, 0.15)	<.001
Group 4	0.17 (0.13, 0.21)	<.001
Group 5	0.16 (0.12, 0.20)	<.001
Sex		
Male	Ref.	
Female	−0.34 (−0.36, −0.31)	<.001
Nonbinary	−0.15 (−1.03, 0.73)	.742
Prefer not to answer	1.10 (0.55, 1.65)	<.001

Key Takeaways

Lessons for Practice

- Spaced repetition enhances metacognitive accuracy: Repeating questions, especially with a two-time spaced strategy, significantly improves learners' confidence in their correct responses, supporting more accurate self-assessment and knowledge retention.
- Confidence gains vary with question type and repetition frequency: Repeated questions yield greater confidence improvements than clone (transfer) questions, suggesting that familiarity reinforces metacognitive alignment more effectively than transfer tasks alone.
- Demographic differences persist but do not diminish intervention effects: While factors such as sex, age, and practice setting influence confidence levels, the positive impact of spaced repetition on metacognitive accuracy remains robust across diverse clinician populations.

Article #5: Tania Jenkins

Social Problems, 2023, **70**, 435–455
doi: 10.1093/socpro/spab046
Advance Access Publication Date: 4 October 2021
Article



Making the Cut: Status, Credentials, and Hiring in Medicine

Tania M. Jenkins 

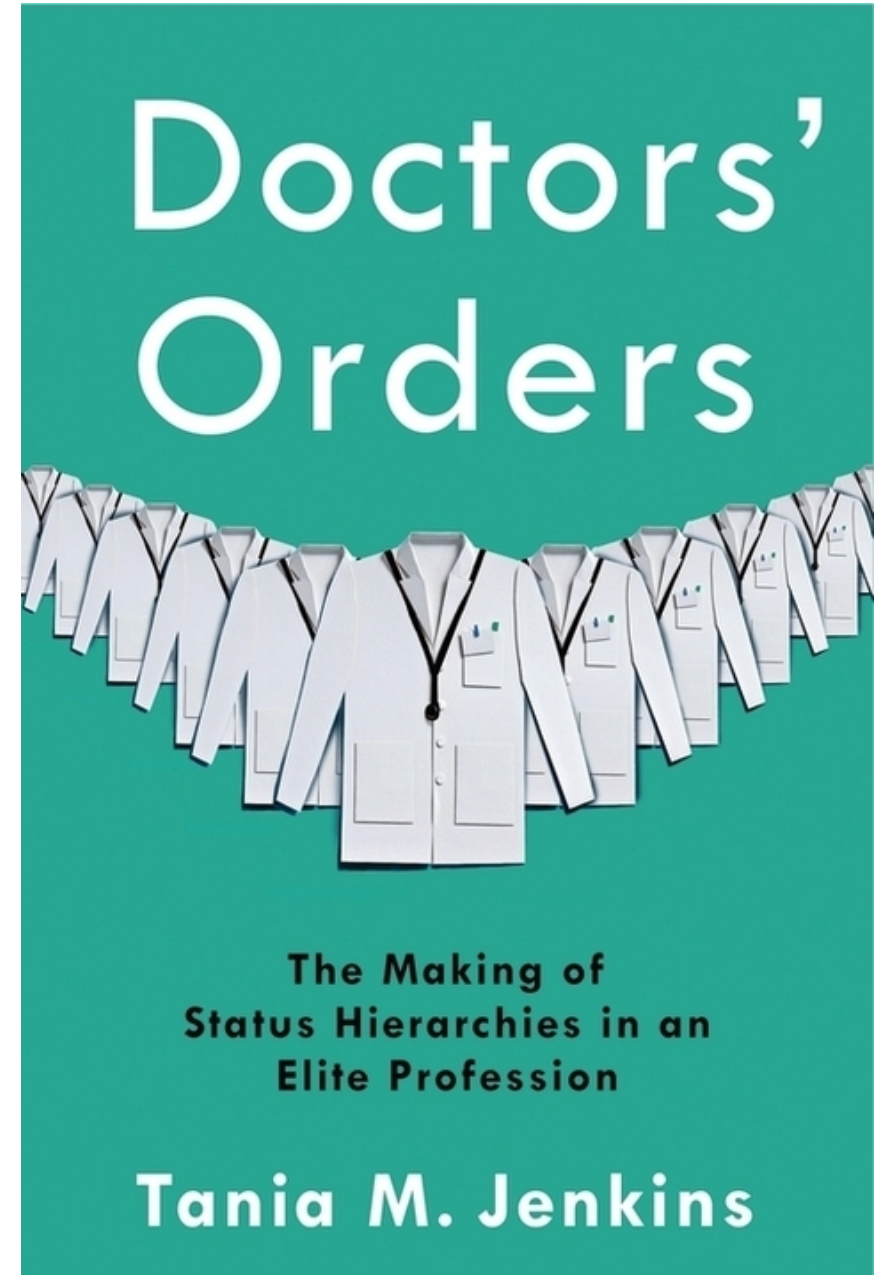
The University of North Carolina at Chapel Hill

ABSTRACT

Little is known about how employers interpret prestigious and non-prestigious credentials in professional recruitment, or how organizations shape and structure this process. Such research is needed to elucidate how employers and organizations contribute to stratification among professionals based on social class, especially since graduates of less selective schools tend to come from lower-SES backgrounds. Using data from a comparative ethnography of residency recruitment at two hospitals, I ask: 1) how do employers interpret prestigious and non-prestigious professional degrees in recruitment, and 2) how does organizational context influence how employers interpret these credentials? The findings add to a growing literature that focuses on qualitative differences between credentials to understand how employers make sense of elite and non-elite credentials within a single profession. The literature disagrees, however, about how employers interpret credentials. I reframe the debate to show that employers rely on combinations of interpretations in sequence, rather than finding evidence for one interpretation or another. I show how organizational context shapes these interpretations, with different employers interpreting the same credentials differently. The article also demonstrates that the order in which employers use different credential interpretations is heavily shaped by organizational status, with implications for inequality.

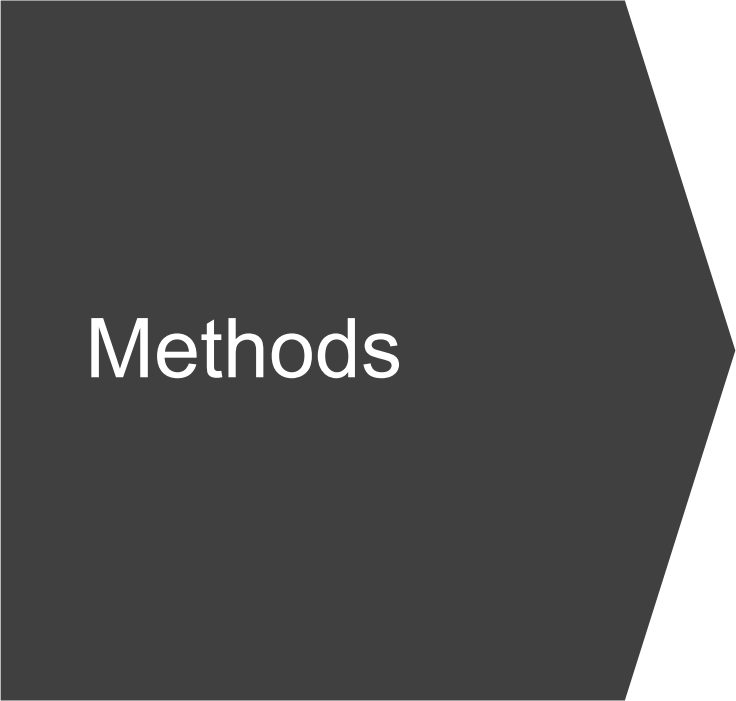
KEYWORDS: hiring; credentials; demand side of hiring; medicine; status.

Read this
book!



The Why

Given widespread segregation along credential lines, and given the considerable overlap among applicants in terms of mean licensing exam scores, internal medicine residency recruitment represents a fruitful case for understanding how employers interpret prestigious and non-prestigious professional credentials in ways that contribute to segregation along degree lines. It also provides a useful case for considering how organizational prestige may play a role in shaping the “Match list,” or labor queue, as employers must be mindful of applicants’ own rank-ordered job queues.



Methods

Embedded Ethnography

- 23 months of fieldwork
- 1,200+ hours of observation
- 123 interviews
- nine focus groups

Two IM Residency Programs

FINDINGS

The internal medicine programs at Community Hospital and University Hospital each received between 2,000–3,000 applications each year. Of these applicants, University Hospital interviewed approximately 500 candidates, while Community Hospital interviewed around 250, for fewer than 100 positions total between the two programs. Importantly, neither program prohibited international or osteopathic applicants from applying. Accordingly, even though Community was lower-tier, it still received hundreds of applications from American-trained MDs, while two-thirds of all applications to University were from non-USMDs.

Their applicant pools had considerable overlap due to the finite number of residency programs, the hospitals' attractive location, the similarity in mean test scores between U.S. and international applicants (NRMP 2018a, 2018b), and the growing number of residency applications per applicant (AAMC 2016). Yet, the house staff in both programs remained highly homogeneous. Community Hospital exclusively matched with non-USMDs, and University Hospital maintained a roster of 98 percent USMDs.¹¹ Self-selection alone cannot explain these segregated outcomes. So how did decision-makers arrive at these decisions?

Embedded Researcher Ethnography

Table 1. Hospital Characteristics: Internal Medicine Residency Programs

		<i>Community Hospital</i>	<i>University Hospital</i>
Number of 3-year residents^a		<i>y</i>	<i>3y</i>
	% USMD^b	0	98
non-USMDs	% USIMG^b	50	0
	% DO^b	11	0
	% IMG	39	2
Nonprofit status		For profit	Not-for-profit
Number of beds^c		<i>x</i>	<i>4.5x</i>
Program's standing (according to respondents)		Non-elite (bottom third of residency programs)	Aspiring elite (top of the second tier / bottom of the first tier program)
% of the medical faculty which were non-USMDs		55	<5
% Residents male		66	60
% Residents non-white		50	25

Note: USIMG = US-citizen international medical graduate; DO = Doctor of Osteopathic Medicine; IMG = International medical graduate

^aTo protect the programs' identity, I do not disclose the actual size of the housestaff.

^bUS citizens

^cTo protect the hospitals' identity, I do not disclose the actual number of beds.

Examples of Themes

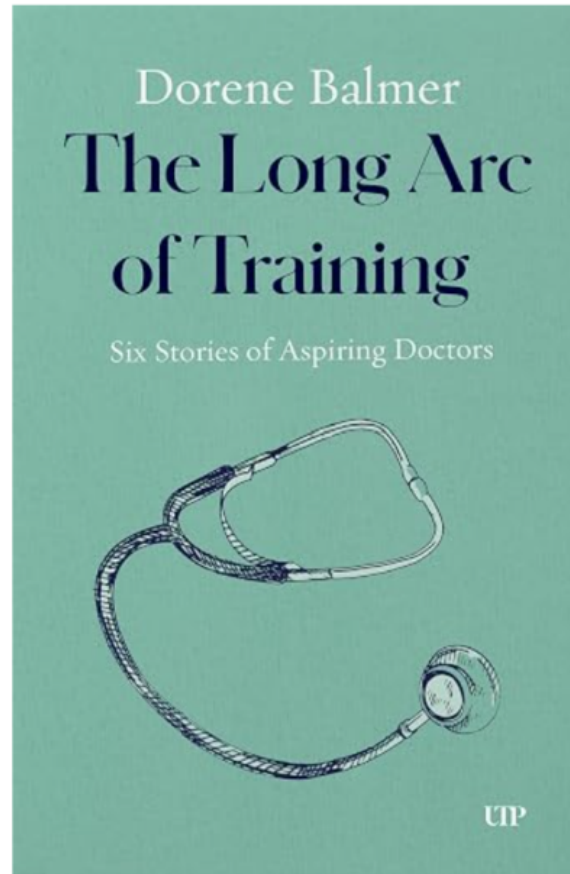
- **Status Hierarchies:** Explores how assumed merit and pedigree influence professional trajectories, effectively sorting doctors into elite and subordinate groups
- **Professional Inequality:** Highlights how international and osteopathic physicians face marginalization, even when outperforming their U.S.-trained peers
- **Medical Sociology Focus:** Provides insights into physician wellbeing, burnout, and job satisfaction, showing how structural and cultural factors in medicine impact both practitioners and patient care

Key Takeaways

“The results suggest that the same credential can hold very different meanings for employers depending on the organization and its status priorities. A Caribbean medical degree, for example, could signal a lack of motivation, as it did at University Hospital, thereby leading decision-makers to exclude that candidate from consideration early in the queuing process. Or it could be interpreted as a sign of grit and determination, a boon to the program’s prestige, as it did at Community Hospital, and therefore come into play after an initial cut based on test scores.

These findings suggest that we should not assume a one-size-fits-all approach to understanding how elite and non-elite employers interpret the same credentials, highlighting the importance of considering organizational context in hiring.”

Read this book, too!



The Long Arc of Training: Six Stories of Aspiring Doctors

by [Dorene F. Balmer](#) (Author) | Format: Hardcover

5.0 ★★★★★ (9)

The Long Arc of Training: Six Stories of Aspiring Doctors follows six medical trainees over twelve years, offering a view into the personal and professional transformations involved in becoming a doctor.

Through a series of interviews conducted during this time, this book captures the deeply human side of medical training. It's not just what these aspiring doctors learn, but how they become who they are.

Unlike memoirs written in hindsight or short-term studies, this book presents longitudinal narratives that unfold gradually. Readers witness how the emotional responses, sense of self, personal well-being, and career plans of doctors-in-the making transform alongside growing medical knowledge and clinical skills.

This book is for anyone curious about the process of becoming a doctor – aspiring medical students, current trainees, practicing physicians, educators, and readers drawn to stories of growth and transformation. With a focus on authentic storytelling, *The Long Arc of Training* challenges common assumptions about medical education and invites reflection on how to best support those in training. It provides a fresh, human-centered perspective on a process often viewed in purely academic terms.

BONUS

Article #6: Pugh et al.

Academic Medicine, 2026, 00, 1–15
<https://doi.org/10.1093/acamed/wvag065>
Advance access publication: March 23, 2026
Review

 **ACADEMIC
MEDICINE** OXFORD
A JOURNAL OF THE AAMC

Association between physician licensing and certification examination performance and quality of patient care: a systematic review

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Abstract

Purpose: Licensure and certification examinations have traditionally served to ensure physicians possess the knowledge and skills required for independent practice. However, with the growing emphasis on workplace-based assessment, the role and relevance of these point-in-time examinations are increasingly debated. This systematic review aimed to explore the association between physician performance on national licensing and certification examinations and subsequent measures of quality of care in practice.

Method: The review followed the Preferred Reporting Items for Systematic Review and Meta-Analyses guidelines. Six databases were searched through November 2025. Original research studies were included if they examined the relationship between performance on medical licensing or certification examinations and physician performance in independent practice, including patient outcomes and fitness to practice. Data were extracted and analyzed using content analysis to identify patterns and trends across studies.

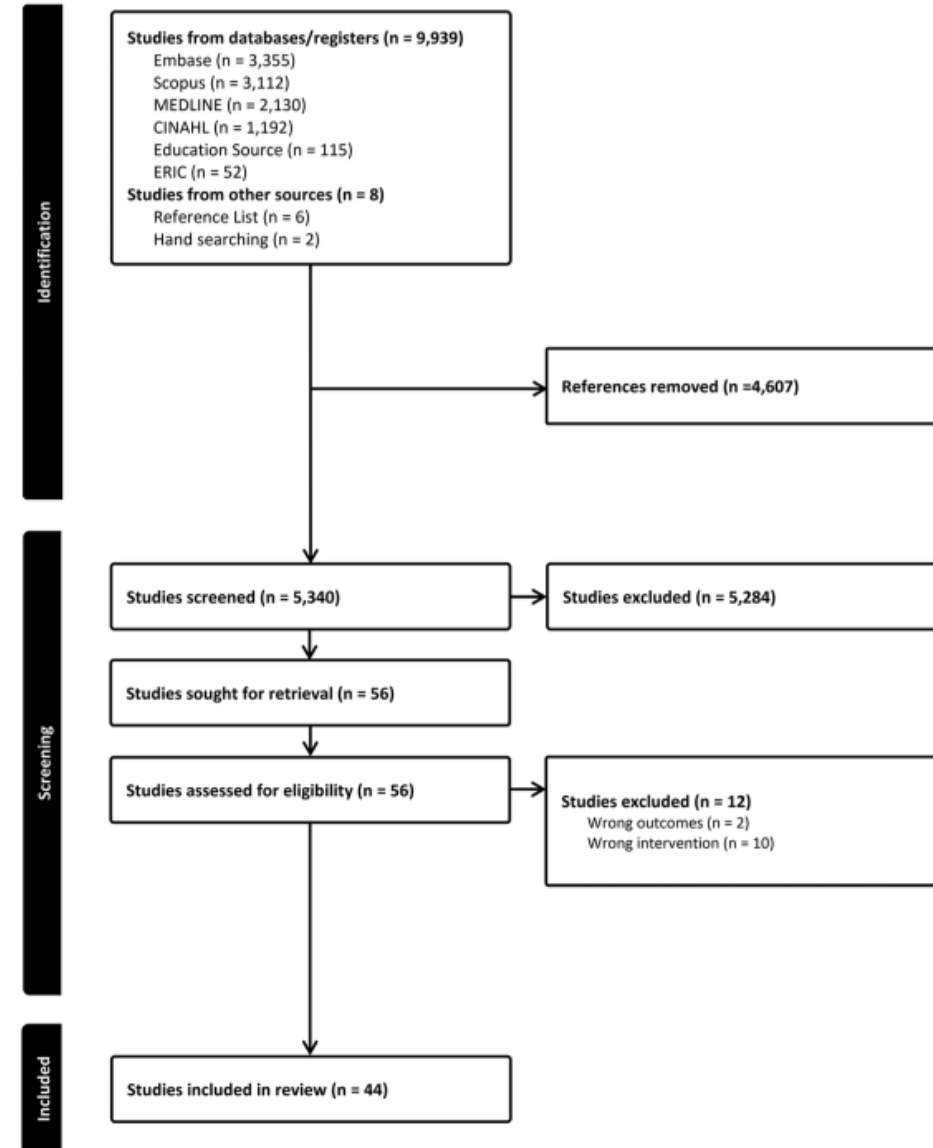
Results: A total of 44 studies involving 1,021,187 physicians were included. The findings demonstrated that examination performance was consistently associated with quality of patient care and fitness-to-practice concerns. Better examination performance was linked to improved adherence to mammography screening recommendations, appropriate prescribing practices, improved care of patients with diabetes, lower patient morbidity and mortality, fewer complaints to regulatory bodies, and lower malpractice payments. The association was observed across examination formats and medical specialties.

Conclusions: The results suggest that national licensing and certification examinations are valuable predictors of future clinical performance. They provide a uniform benchmark to ensure physicians meet rigorously defined criteria. The findings support continued use of these examinations as part of a comprehensive assessment program to promote high-quality patient care and physician competence. This is critical in an era of increased physician mobility, where many physicians have trained in jurisdictions with widely varying curricula and assessment practices. National examinations help ensure that all licensed physicians meet a consistent standard for clinical competence.

The Why

- Association between physician's performance on national licensing and certification examinations and subsequent measures of quality of practice
- Systematic review of literature
- Outcomes related to either quality of patient care, or fitness to practice

44 studies
included
encompassing
1,021,187
physicians



Examples of Themes

- Adherence to screening recommendations
- Prescribing practices
- Morbidity/mortality
- Diabetes care
- Length of hospital stay
- Complaints to regulatory boards
- Malpractice claims

Key Takeaways

- Higher scores:
 - Superior clinical outcomes
 - Lower patient mortality rates
 - Lower risk of subsequent state medical board disciplinary actions, regulatory complaints, or medical malpractice payments
- Validation of point-in-time, standardized examinations as reliable, long-term indicators of a physician's real-world clinical competence, safety, and professional accountability



Article #7: Merkebu et al.

Emotions and Clinical Reasoning in Medical Education and Clinical Practice: A Scoping Review

Jerusalem Merkebu, PhD, Michael Y. Soh, PhD, Michael Loncharich, MD, Matthew K. Hawks, MD, MHPE, Joseph A. Costello, MSIS, Marina Shapiro, PhD, Lauren A. Maggio, PhD, Steven J. Durning, MD, PhD, and Binbin Zheng, PhD

Abstract

Purpose

Clinical reasoning, an essential skill in medicine, is significantly influenced by emotions. Although the cognitive aspects of clinical reasoning have been extensively studied, the impact of emotions remains underexplored. This scoping review aimed to map the existing literature on the influence of emotions on clinical reasoning across medical education and practice.

Method

The authors conducted a scoping review (March 11, 2022, and December 1, 2023) following Arksey and O'Malley's framework of MEDLINE (Ovid), Embase (Elsevier), Web of Science, and PsycInfo (Ebsco) to identify relevant literature. All empirical studies examining the interplay of emotions and clinical reasoning on the continuum of medical education training were eligible for inclusion.

Results

Sixty-one studies met the inclusion criteria. The findings indicate that negative and anxiety-provoking emotions, such as stress, fear, and uncertainty, are predominantly highlighted in the literature as influencing clinical reasoning. These emotions were associated with conservative clinical approaches, increased diagnostic errors, and suboptimal management decisions. Anxiogenic emotions showed mixed effects, sometimes leading to more thorough evaluations but also potentially prompting unnecessary interventions. Conversely, positive emotions and gut feelings, often cultivated through experience, were found to support effective clinical reasoning and potentially improve patient care by fostering adaptability and diagnostic accuracy.

Conclusions

This review captures the integral role of emotions in clinical reasoning, impacting every facet of patient care. The nuanced interplay between emotions—both positive and negative—and clinical reasoning emphasizes that emotions not only arise as by products of clinical encounters but also actively shape clinical reasoning. These findings call for educational interventions that equip clinicians with emotional awareness and regulation strategies to enhance their clinical reasoning. Future research should explore the influence of positive emotions, investigate the effectiveness of emotional regulation interventions, and address gaps in understanding the contextual and cultural variability in emotional dimensions of clinical reasoning.

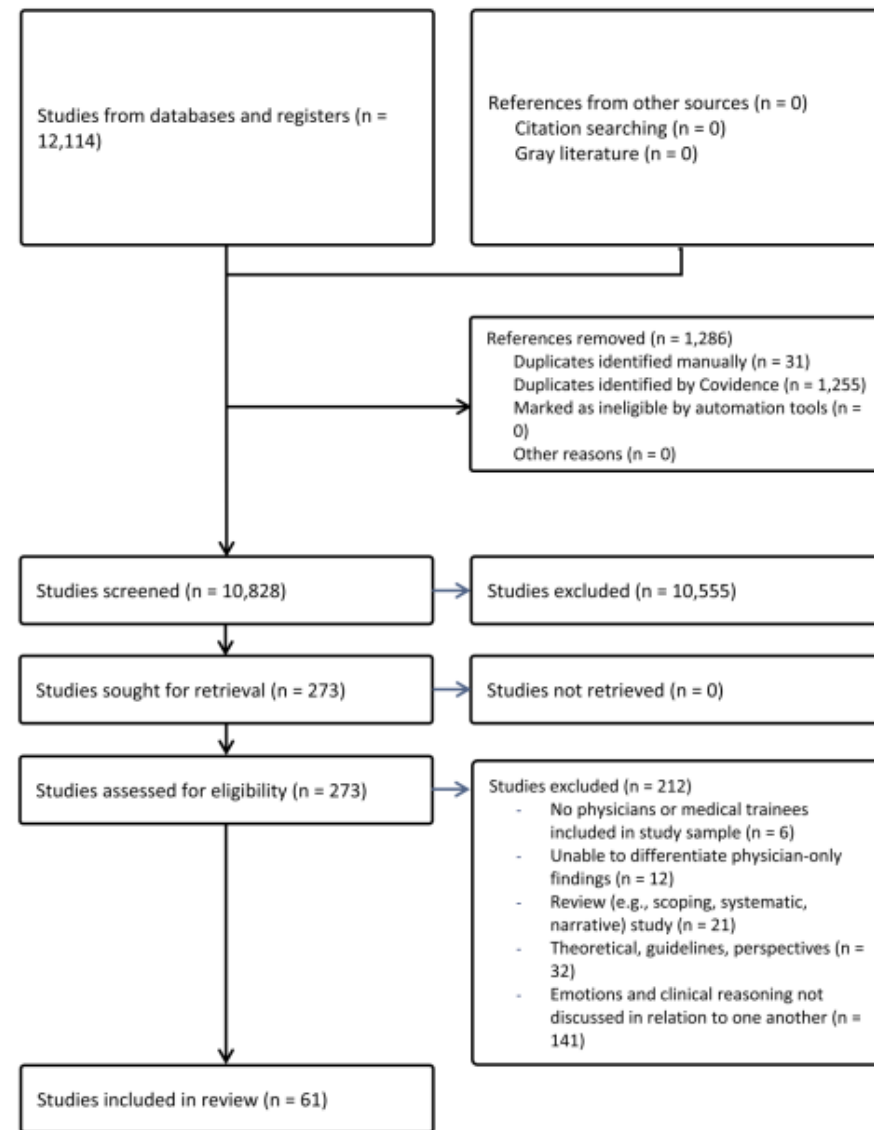
The Why

- The impact of emotion on clinical reasoning across medical education and practice
- The idea that **emotions are not mere noise** that occasionally disrupt clinical care; rather, they are **active components of clinical judgment** that directly shapes how medical decisions are made

The How

- Scoping review between March 11, 2022, and December 1, 2023
- Included all peer-reviewed, empirical studies that examined the direct interplay of emotions and clinical reasoning across the continuum of medical education, training, and practice
- 61 studies that met eligibility requirements

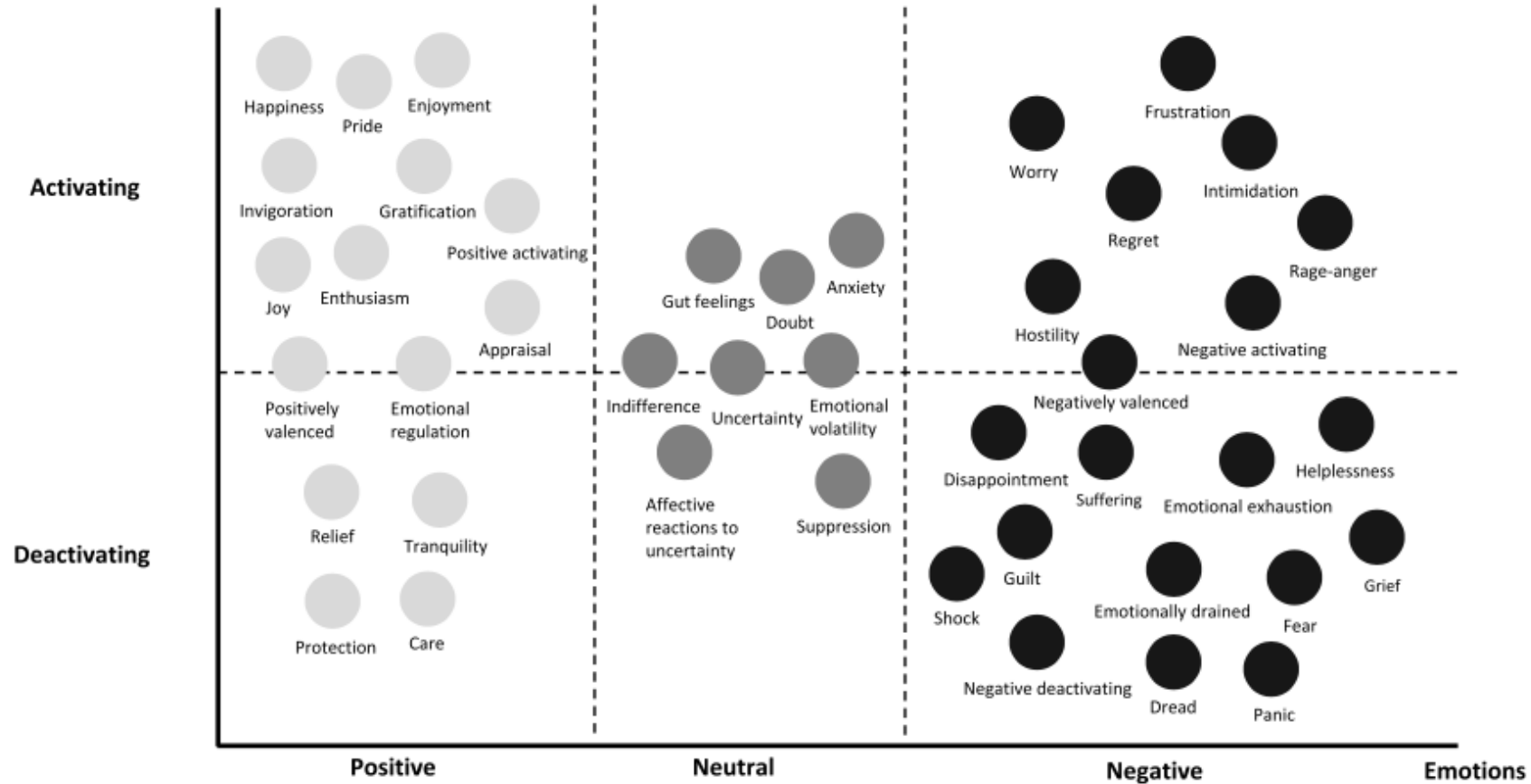
Methodology



Types of Factors Triggering Emotions

- Three levels:
 - Physician-level
 - Patient-level
 - System-level
- Activating and deactivating:
 - Positive
 - Neutral
 - Negative

The Findings



The Key Takeaways

- Complex interaction between emotion and cognition in clinical settings
- Emotions significantly influence diagnostic accuracy, cognitive load, and the overall quality of patient care
- **Implications...** Highlights a need for medical curricula to incorporate emotional awareness and regulation training
- **Next steps...** and the assessment...?

Article #8: Kumbamu et al.

Academic Medicine, 2026, 00, 1-9
<https://doi.org/10.1093/acamed/wvag211>
Advance access publication: July 8, 2026
Research Report

 **ACADEMIC
MEDICINE** OXFORD
A JOURNAL OF THE AAMC

Cultivating caring practices in US clinical learning environments: a multi-site qualitative exploration of actions, barriers, and strategies

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Abstract

Purpose: Health care professionals perceive caring as fundamental to their practice, viewing it as both a calling and a virtue. Yet, the literature on caring underscores a crisis of genuine commitment to caring and a growing need to foster caring values and actions within clinical learning environments. This study aimed to explore caring actions; examine perceived systemic, institutional, and individual barriers and facilitators to caring in health care; and identify multilevel strategies to promote caring behaviors in the clinical learning environment.

Method: Researchers conducted 15 focus groups with 50 participants across 6 medical schools in the United States, including physicians, residents, medical students, nurses, administrative staff, and patients. A general inductive qualitative approach was used to explore the complexities and contradictions in caring values, strategies, and actions. Data were collected and analyzed from April 2021 to October 2023.

Results: Results revealed actions that promote patient-centered care (eg, caring gestures, trust-building), support health care teams (eg, recognizing good work), nurture learners (eg, promoting reflection through feedback), and enhance individual well-being (eg, taking personal time). Participants also identified barriers to caring across individual (eg, burnout and exhaustion, cultural and language barriers, lack of self-care), institutional (eg, lack of institutional support, limited time, protocols, and processes), and systemic (profit-driven health care, systemic pressures, health disparities, administrative burdens) levels in the clinical learning environment.

Conclusions: Caring within clinical learning environments is not merely an individual concern, but also an interaction among systemic, institutional, and individual dimensions that can impact and conflict with one another. Findings can inform the development of structured frameworks that effectively integrate caring values into medical curricula, thereby promoting the education of health care professionals who are both clinically competent and committed to providing compassionate care to patients, co-workers, learners, and themselves.

The Why

- About: caring and compassion in clinical environments
- Identify systemic, institutional, and individual barriers to caring
- Identify strategies to reinforce compassionate practices

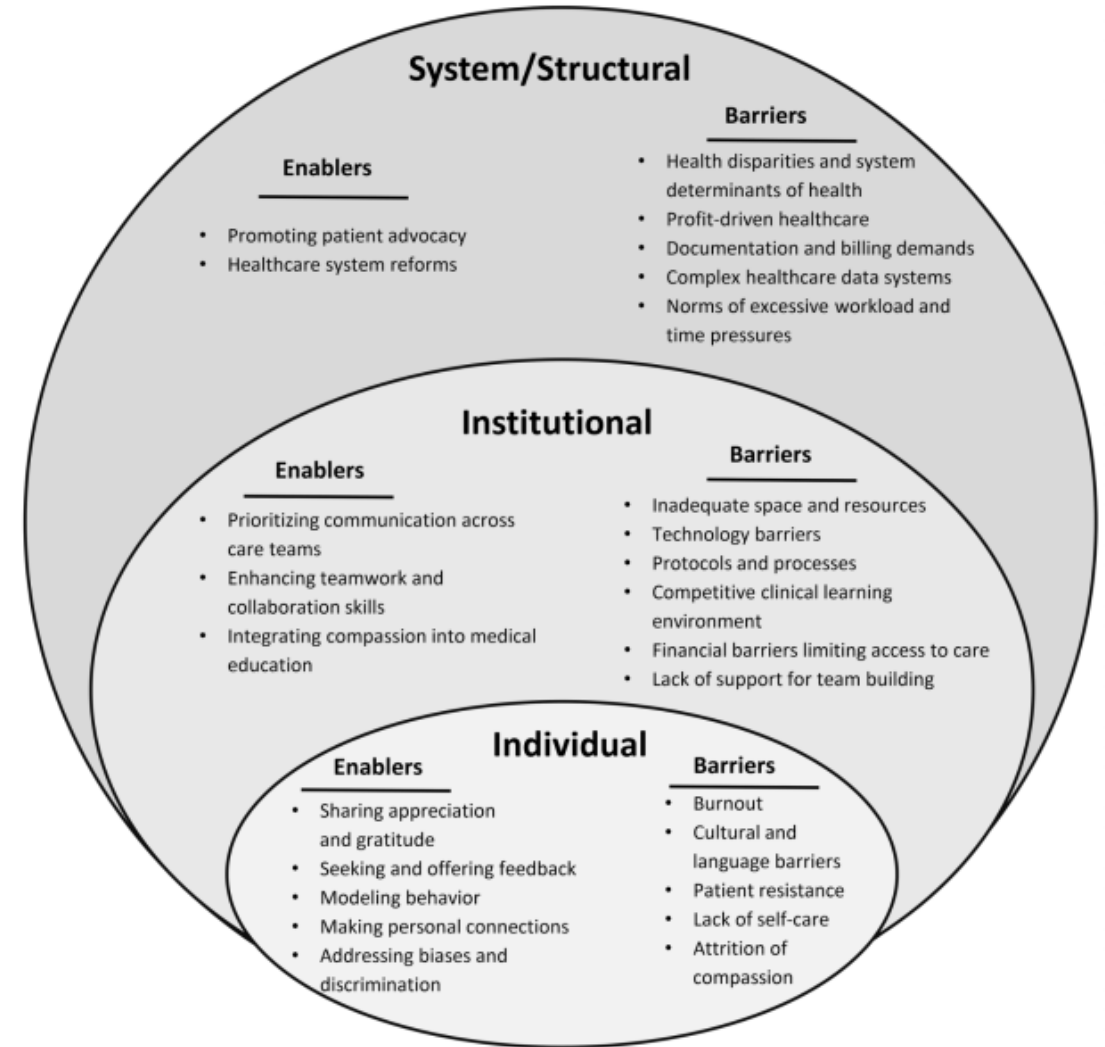
The How

- Multi-site research across six US medical schools
- Included physicians, residents, medical students, nurses, administrative staff, and patients
- 15 focus groups; 50 total participants
- April 2021 to October 2023
- Qualitative analysis of data

Three themes

- Caring behaviors/actions for:
 - Patients
 - Teams
 - Learners
 - Self
- Barriers (institutional, individual, systemic)
- Strategies for support (institutional, individual, systemic)

Systemic, Institutional, and Individual Barriers to Caring



Key Takeaways

- Caring should not be treated merely as a personal responsibility
- Compassion in healthcare is an ecosystem shaped by the overlapping (conflicting?) forces of systemic, institutional, and individual dimensions
- Curriculum integration?
- **Next steps...** assessment integration?

Article #9: Schumacher et al.

Advancing, Graduating, and Attesting Readiness of Pediatrics Residents With Concerns

Daniel J. Schumacher, MD, PhD,¹ Benjamin Kinnear, MD, MEd,^{1,2} Patricia Poitevien, MD, MSc,³ Robert Daulton, MD,¹ Ariel S. Winn, MD⁴

abstract

OBJECTIVE: Determine the prevalence of advancing and graduating residents with concerns as well as attesting readiness to the American Board of Pediatrics for residents not deemed ready at graduation among pediatric program directors.

METHODS: Surveys were sent to pediatric program directors from July to September 2024 to explore their experiences of graduating residents for whom they had concerns, graduating residents they would not trust to care for their loved ones, attesting readiness to the American Board of Pediatrics despite concerns, and advancing residents from 1 year to another despite having concerns.

RESULTS: 117/208 (56%) program directors completed the survey. A total 79% (93/117) reported graduating at least one resident despite having concerns about doing so. Fewer (43%, 50/117) had graduated at least one resident they would not trust to care for their loved one. Fewer, 26% (31/117), reported attesting to readiness for a resident to sit for the American Board of Pediatrics initial certification exam despite concerns regarding the resident's ability to practice without supervision as a general pediatrician. Finally, 79% (92/117) reported advancing at least one resident from one training year to the next despite concerns regarding the resident's ability to take on additional responsibility.

CONCLUSION: These findings add to a growing literature raising concerns about the readiness of some residency graduates to meet the needs of patients. While program directors served as the source of information in this study, collective responsibility for both the issues presented and the ownership in addressing them are wide ranging.



WHAT'S KNOWN ON THIS SUBJECT: Gaps in readiness at the completion of training have been reported for residents in multiple specialties and countries.

WHAT THIS STUDY ADDS: Advancing and graduating pediatrics residents with concerns and attesting readiness to the American Board of Pediatrics when residents may not be ready are commonplace, continuing to heighten concerns about readiness of residency graduates to meet the needs of patients.

Article #10: Koempel et al.

A Qualitative Study of Family Physician and Patient Perspectives on Professionalism

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<https://doi.org/10.3122/jabfm.2026.260058R0>

Journal of the American Board of Family Medicine

Vol. 39, Issue 1, 2026

Despite a robust literature on the topic, a consistent definition of medical professionalism remains elusive. A single, generalizable definition is ineffective and tends to emphasize prohibited behaviors rather than aspirational concepts. As physicians are increasingly employed by corporate entities, the nature of healthcare work has changed, as well as professional expectations. This study explores how patients and family physicians understand medical professionalism and its drivers.

We recruited family physicians who completed the ABFM Continuing Certification Questionnaire in 2024 to participate in in-depth, semi-structured interviews. A trained qualitative researcher used Zoom to interview 27 demographically diverse physicians. We utilized modified snowball sampling to recruit patients through a key informant. All interviews were transcribed verbatim and analyzed utilizing NVivo software following an inductive content analysis approach.

Interview results coalesced into a model of professionalism that represents the clinic visit, the practice site, and the overarching system. Three main components of individual physician professionalism emerged, including foundational duties, relationship-based care, and communication. Patient trust in the overarching healthcare system and individual physicians was deeply influenced by the interconnected system.

This study presents key elements of physician professionalism while highlighting how deeply physicians are entangled within the larger practice and healthcare systems. Professionalism extends beyond—and is impacted by—more than individual attitudes and behaviors. Institutions must interrogate their own policies to support physicians in developing their skills in relationship-based care and communication.

The Why

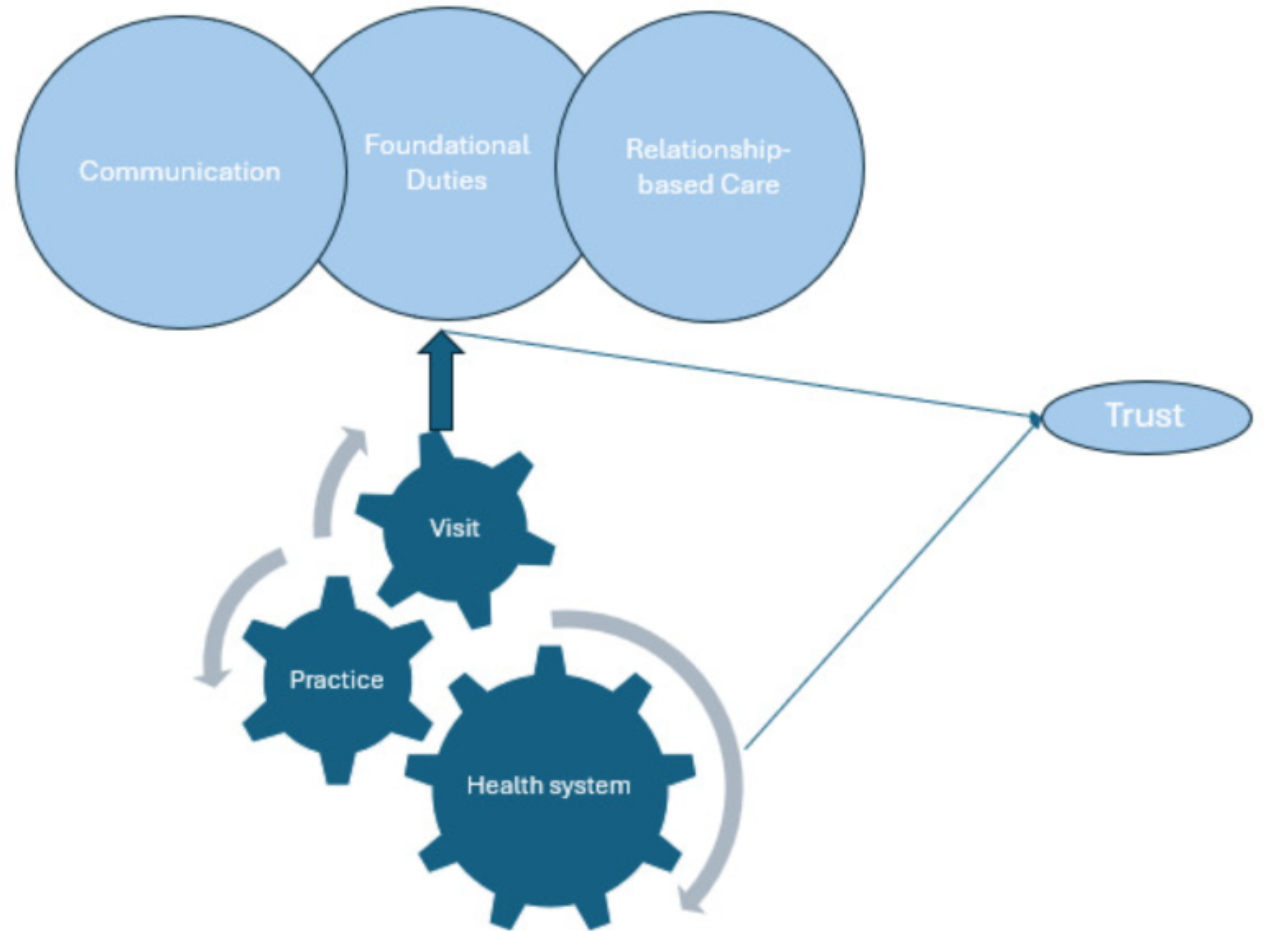
- What defines **professionalism** for a physician?
 - Historically seen as what “not” to do instead of what needs to be achieved (prohibited vs. aspirational behavior)
- Explores how patients and family physicians understand medical professionalism

The How

- Random sample of 325 ABFM diplomates invited
- Final sample of 27 physicians and 11 patient advisors
- Gather integrated viewpoints on professionalism
- Sem—structured remote interviews
- Inductive content analysis approach

The Framework

Figure 1. Model of professionalism developed from 27 physician and 11 patient advisor interviews, 2025.



The Key Takeaways

- Several intertwined components
- Requires a multi-pronged approach spanning visits, sites, and the health system

Lessons

- Certification continues to be associated with downstream benefits (Seguchi, Gray, van der Grift)
 - More complex assessments of motivations could be beneficial (Gavarkovs, Koempel)
 - Social structures bear examination (Jenkins, Balmer)
 - Proliferation of excellent methods in certification research including 12 year long ethnographies and increased use of registries.
- 